PLTW POE

Problem 1.3.1 Fix Fran’s Farm

PLTW

FIX FRAN’S FARM

Lesson 1 of 3

No Rest for the Weary

GOALS	MATERIALS	RESOURCES

- Apply knowledge of mechanical design, simple machines, and mechanisms to an agricultural design context.

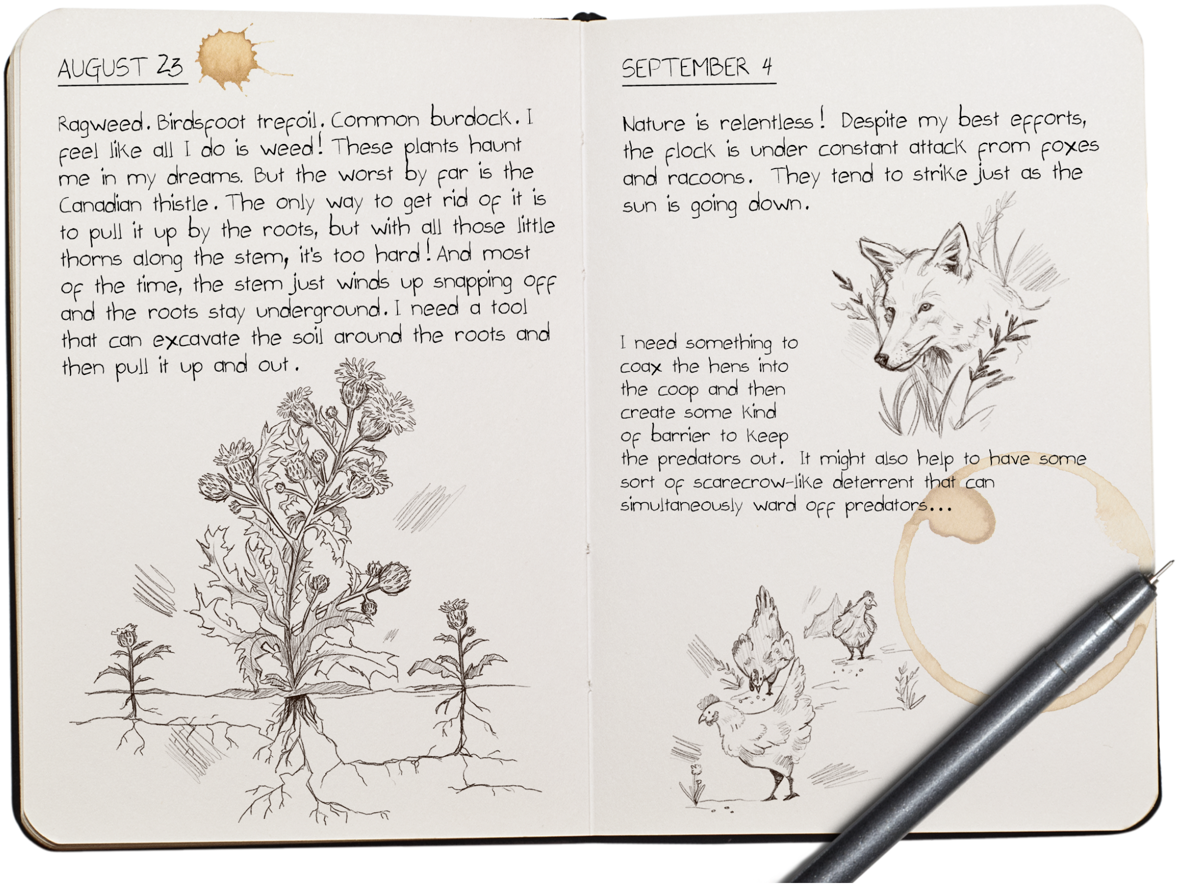
GOALS	MATERIALS	RESOURCES

- VEX® V5 POE Kit
- Additional items as needed for the end task

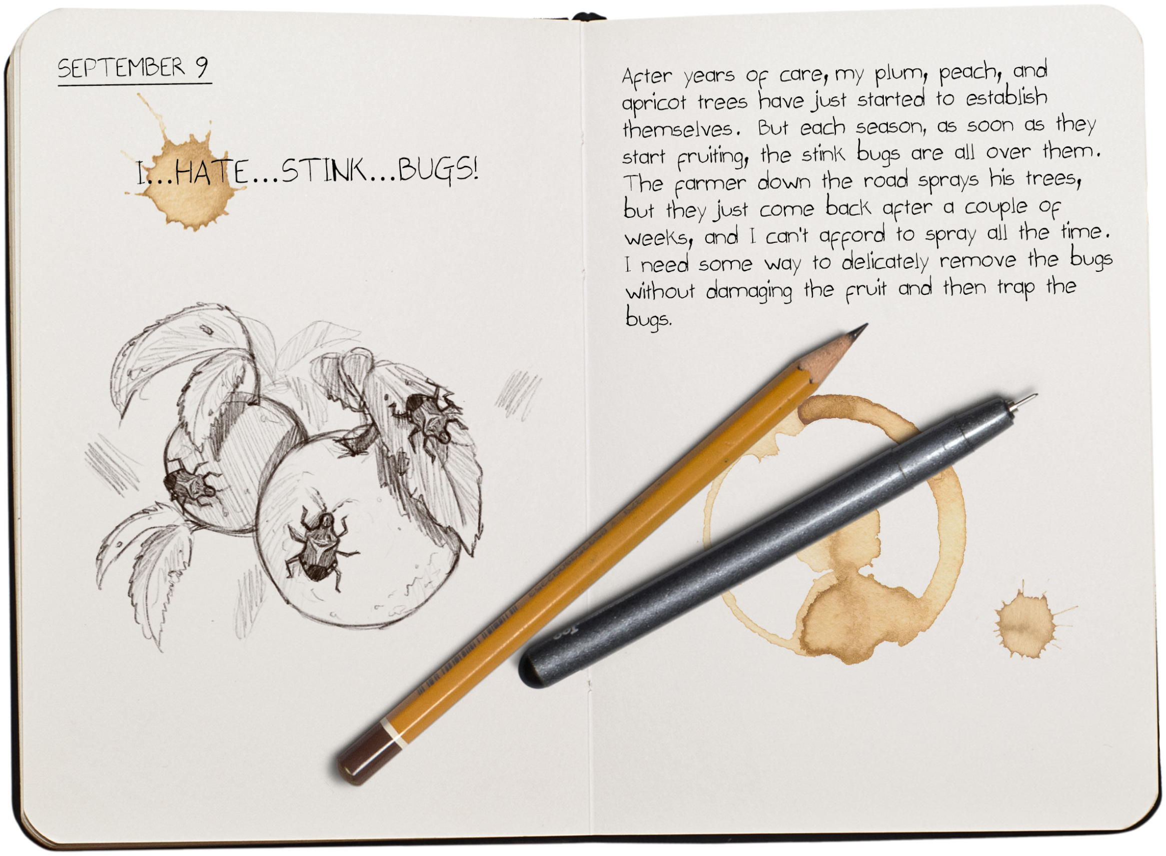
GOALS	MATERIALS	RESOURCES

- **Fran’s Journal Entries**
- **Decision Matrix Template**
- **Problem 1.3.1 Fix Fran’s Farm Rubric**
- **Citing Sources**
- **Technical Writing**
- **VEX® Parts**
- **VEX® V5 Build Support Guides**
- **Problem 1.3.1 Fix Fran’s Farm (Downloadable PDF)**

Thanks in part to your efforts, Fran’s farm is now a profitable business! You’ve helped her along the way by designing a latch mechanism that helps protect her flock of chickens and an egg grabber for her coop. You also helped her understand how simple machines can be used to make her life easier. But with a farm, there are always new challenges and new opportunities. Fran diligently records some of the issues she faces in a journal. Review Fran’s two journal entries in the Journal Entries presentation. (You can also download **Fran’s Journal Entries**.)



Journal Entries for August 23 and September 4



Journal Entry for September 9

Journal Entries presentation

In teams, select one of Fran's pressing problems as described in the journal entries. Work through the design process and apply your knowledge of mechanical engineering to develop a viable solution to the problem that you select. Your solution must meet the following criteria:

- A single human input force
- A minimum of three mechanisms

As in the optional unit's problem, you will document your design process. Upon completion of the design process, you will pitch your solution to your classmates who will listen to your pitch as though they were **angel investors**.

Deliverables and Project Portfolio

1. **Problem Definition**. Fully articulate Fran’s problem objectively and succinctly.

2. **Brainstorm**. With your team, develop at least three potential solutions to your chosen problem. Conduct research to help spark ideas.

> **Citing Sources**: For additional support, see **Citing Sources**.

3. **Decision Matrix**. Complete a **Decision Matrix Template** to determine the best solution.

4. **Technical Drawing.** Create technical drawings that effectively model your three mechanisms.

5. **Test Development**. Design a method to test the **efficacy** of your design solution.

6. **Prototype 1**. Construct a prototype of your design solution that can be tested.

NOTE: If you need help while building, reference **VEX® Parts** and **VEX® V5 Build Support Guides**.

7 **Test and Evaluate**. Conduct a test of your prototype.

8 **Peer Review**. Present your design solution to another group to receive feedback. Also, review another team's design and provide feedback.

9 **Revise Design**. Iterate your design based on the evaluation of your test results and peer feedback.

10 **Prototype 2**. Modify your prototype based on your revised design.

11 **Evidence**. Perform calculations of each of your mechanisms to prove the mechanical advantage your device creates.

12 **Investor Pitch**. Using the information compiled in your portfolio, prepare and deliver an investor pitch to your classmates.

Refer to **Problem 1.3.1 Fix Fran's Farm Rubric** for success criteria for each deliverable.

Unit Reflection

Use the following prompts to write a reflection of your experience in the unit problem. Use the **Technical Writing** resource to help write your narrative.

Question 1

To what extent does your design solution solve Fran's problem? Support your claim with evidence from your testing and evaluation. While this was a contrived scenario, are there merits to your design, or is this a practical solution to the problem as described (meaning is there a marketable idea here)? Why or why not?

Question 2

How much did your design change through iteration? If given another opportunity to iterate, what might you do differently? Why?

Question 3

Through this course so far, you participated in several full design cycles. Is there any part of the process that you enjoyed most? What type of engineer is most likely to perform that part of the process on a routine basis?

Question 4

Conduct web research to find a current job posting for that specialization. Does the job description align with your understanding of that specialization? Does it look more or less interesting to you than you expected? Explain your reasoning.